

DYLAN NAGEL

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EDUCATION

University of Waterloo

Sep. 2023 – Expected 2027

Bachelor of Computer Science – Honours Cooperative Program

Waterloo, ON

- **Cumulative GPA:** 3.9 | **President's Scholarship of Distinction** | **Coursework:** Application Development, Algorithms, Data Structures & Data Management, User Interfaces, Operating Systems

EXPERIENCE

Software Engineering Intern

May 2026 – Aug. 2026

Apple

Cupertino, CA

- **Sole engineer** on a **prompt and agent evaluation platform** shipped end to end across 9 Django apps (**React 19/TypeScript, Django Ninja, Celery/Valkey**), scoring single-turn prompts on **labeled datasets** and **agentic prompts on full tool-call trajectories**; raised vulnerability **CWE classification accuracy from 0.49 to 0.95**.
- Shipped the **LLM assistant layer** for the security organization's vulnerability dashboard, where a plain-English request rewrites filters, rearranges panels, or mutates records through **47 typed tools**, each user-confirmed and **RBAC-checked** at execution; tested with **pytest** and **Playwright**, and adopted by **4+ internal applications**.
- Contributed to the security organization's work on **Project Glasswing**, **Anthropic's** early-access program for its pre-general-availability **Mythos** model.
- **Self-identified** and fixed a bottleneck where **14 paginated endpoints** across **7 Django apps** loaded **entire tables into memory**; deferring pagination to **Postgres** cut production ticket list load times **from over 5s to under 300ms**.

Software Development Intern

Sep. 2025 – Dec. 2025

Carta Inc.

Waterloo, ON

- Engineered an AI email routing service (**Python/Django**) that classifies and routes **1.5M annual emails**, projected to handle **15%** without human triage and cut six figures of yearly operating cost.
- Designed the service's **RESTful API** and refactored the **PostgreSQL schema** to **log model predictions** and surface them in the dashboard.
- Enhanced the Fund Admin dashboard, scaling to serve over **2,000 monthly business users**, by implementing React features and integrating Snowplow for analytics within a **Docker** and **Kubernetes** CI/CD pipeline.

Full Stack Developer

May 2024 – May 2025

Empathia.ai

Pasadena, CA (Remote in Canada)

- Developed full-stack **RAG** features (**Python/TypeScript**) that leveraged user notes to boost AI smart-edit accuracy by **30%** for over **1,000 users**.
- Launched a blog portal (images on **S3**, rich text in **MongoDB**) enabling autonomous publishing of SEO-rich posts with metadata, growing organic traffic **20%**.
- Owned Empathia.ai's marketing site end to end (**React, Node.js**), including build, deploy, and analytics instrumentation, driving **68% search traffic growth** in a single quarter.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C, C++, C#, SQL, Bash

Frontend: React, Zustand, TanStack Query/Router, Vite, styled-components, Tailwind, HTML/CSS

Backend: Django, Django Ninja, Celery, Node.js, Express, FastAPI, REST APIs, SSE, .NET

Databases & Cloud: PostgreSQL, MongoDB, SQLite, Redis/Valkey, AWS (S3), schema design

Testing & DevOps: pytest, Playwright, Vitest, Docker, Kubernetes, GitHub Actions, CI/CD, Git

PROJECTS

🔗 Acronymize | *React, TypeScript, Python, Django, PostgreSQL, Git*

- Created a full-stack word puzzle game with a **React/TypeScript** frontend and **Django** backend, with three game modes and authenticated per-user progress tracking.
- Implemented Wordle-style positional hints alongside real-time guess similarity scoring, served from the Django backend by an **ML sentence transformer** model over cached puzzle embeddings.

🔗 Breadcrumbs | *Kotlin, Jetpack Compose, Mapbox SDK, Supabase (Postgres/PostGIS), Git*

- Designed the **Postgres/PostGIS** schema and **Supabase RPC** query layer powering an interactive **Mapbox** trip map, enforcing **row-level security** scoped to trip membership and an accepted friend graph, with debounced/batched viewport queries to keep continuous pan/zoom from flooding Postgres.